Лабораторна робота №7

«Collatz's hypothesis»

«Some actualevaluations – finding the value of π»

«Drawing squares (actually: rectangles) »

група ІСТ-21

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Варіант: 18

Lab 2.3.1

```cpp
#include <iostream>

using namespace std;

int main() {
    int c0;
    cin >> c0;

    if (c0 <= 0) {
        return 0;
    }

    int steps = 0;

    while (c0 != 1) {
        if (c0 % 2 == 0) {
            c0 = c0 / 2;
        } else {
            c0 = 3 * c0 + 1;
        }
        cout << c0 << endl;
        steps++;
    }

    cout << "steps = " << steps << endl;

    return 0;
}
```

```
16
8
4
2
1
steps = 4
PS D:\C++\Lab7>
```

Lab 2.3.2

```cpp
#include <iostream>

using namespace std;

int main(void) {
    double pi4 = 0.;
    long n;

    cout << "Number of iterations? ";
    cin >> n;

    for (long i = 0; i < n; i++) {
        if (i % 2 == 0) {
            pi4 += 1.0 / (2 * i + 1);
        } else {
            pi4 -= 1.0 / (2 * i + 1);
        }
    }

    cout.precision(20);
    cout << "Pi = " << (pi4 * 4.) << endl;
    return 0;
}
```

```
Number of iterations? 10000
Pi = 3.1414926535900344895
PS D:\C++\Lab7>
```

Lab 2.3.5

```cpp
#include <iostream>

using namespace std;

int main(void) {
    int n;
    cin >> n;

    if (n <= 1) {
        cout << "Sorry, the side size is too small" << endl;
        return 0;
    }
    if (n > 20) {
        cout << "Sorry, the side size is too big" << endl;
        return 0;
    }

    cout << '+';
    for (int i = 0; i < n - 2; i++) {
        cout << '-';
    }
    cout << '+' << endl;

    for (int i = 0; i < n - 2; i++) {
        cout << '|';
        for (int j = 0; j < n - 2; j++) {
            cout << ' ';
        }
        cout << '|' << endl;
    }

    cout << '+';
    for (int i = 0; i < n - 2; i++) {
        cout << '-';
    }
    cout << '+' << endl;

    return 0;
}
```

```
4
+--+
|  |
|  |
+--+
PS D:\C++\Lab7>
```